$1\underbrace{0\ \dots\ 0}_{n}{}_{(2)} = 2$^n

$\underbrace{1\dots\ \dots\ 1}_{n}{}_{(2)} = 1\underbrace{0\dots\dots0}_{n}{}_{(2)}\ \ -1_{(2)} =$ 2^(n-1)

Ex1. Subunitary numbers – codes and operations in complementary code

n=8 bits

6/8=6*8^(-1)= 0,6(8)=0,110(2)

X= 11/16 = 11*16^(-1)= 0,B(16)=0,1011(2)
Y= 0,45 = 0,346(8)=0,011100110(2)
0,45*8=3,6
0,6*8=4,8
0,8*8=6,4

positions	S 7 ,	6	5	4	3	2	1	0
$[11/16]_{dir} = [11/16]_{inv} = [11/16]_{compl} =$	0	1	0	1	1	0	0	0
$[-11/16]_{dir} =$	1	1	0	1	1	0	0	0
$[-11/16]_{inv} =$	1	0	1	0	0	1	1	1
$[-11/16]_{compl} =$	1	0	1	0	1	0	0	0

positions	S 7 ,	6	5	4	3	2	1	0
$[0,45]_{dir} = [0,45]_{inv} = [0,45]_{compl} =$	0	0	1	1	1	0	0	1
$[-0,45]_{dir} =$								
$[-0,45]_{inv} =$								
$[-0,45]_{compl} =$	1	1	0	0	0	1	1	1

$[11/16+0,45]_{compl} = [11/16]_{compl} \oplus [0.45]_{compl}$

		S ,									
$[11/16]_{compl} =$		0	1	0	1	1	0	0	0	$\oplus$	**Overflow (r1)**
$[0,45]_{compl} =$		0	0	1	1	1	0	0	1		
		1	0	0	1	0	0	0	1		

$[11/16- 0,45]_{compl} = [11/16]_{compl} \oplus [-0.45]_{compl}$

		S ,									
$[11/16]_{compl} =$		0	1	0	1	1	0	0	0	$\oplus$	**Correct result (r2)**
$[-0,45]_{compl} =$		1	1	0	0	0	1	1	1		**[2^(-3)+2^(-4)+2^(-5)+**
	~~1~~	0	0	0	1	1	1	1	1		**2^(-6)+2^(-7)]compl**

$[0{,}45-11/16]_{compl} = [0{,}45]_{compl} \oplus [-11/16]_{compl}$

		S ,									
$[-11/16]_{compl}$		**1**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	$\oplus$	**Correct result**
$[0{,}45]_{compl}$ =		**0**	**0**	**1**	**1**	**1**	**0**	**0**	**1**		**[-(2^(-3)+2^(-4)+2^(--5)+ 2^(-6)+2^(-7))]compl**
		1	**1**	**1**	**0**	**0**	**0**	**0**	**1**		
complement		**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**		

<u>Ex2</u>: Represent in floating point notation, single precision (SP), with mantissa>1, the number 2053,45

2053= 2048+4+1= 2^11+2^2+2^0=100000000101 (2)

0,45= (8)= (2)

Y= 0,45 = 0,3463(8)=0,011100110011(2)

0,45*8=3,6
0,6*8=4,8
0,8*8=6,4
0,4*8=3,2

2053,45= 100000000101, 011100110011 (2)=

=1,00000000101011100110011 (2)* 2^(11), e=11

Mantissa

1 –hidden bit (it is not represented internally)

c=127+e=127+11=128+8+2=2^7+2^3+2^1=10001010(2)

	8 bits								23 bits																						
S	c =127+e							→ , ←				mantissa																			
0	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	1	0	0	1	1	0	0	1	1
4				5				0				0				5				7				3				3			